DynaGuide: Steering Diffusion Polices with Active Dynamic Guidance

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Abstract

Deploying large, complex policies in the real world requires the ability to steer them to fit the needs of a situation. Most common steering approaches, like goal-conditioning, require training the robot policy with a distribution of test-time objectives in mind. To overcome this limitation, we present DynaGuide, a steering method for diffusion policies using guidance from an external dynamics model during the diffusion denoising process. DynaGuide separates the dynamics model from the base policy, which gives it multiple advantages, including the ability to steer towards multiple objectives, enhance underrepresented base policy behaviors, and maintain robustness on low-quality objectives. The separate guidance signal also allows DynaGuide to work with off-the-shelf pretrained diffusion policies. We demonstrate the performance and features of DynaGuide against other steering approaches in a series of simulated and real experiments, showing an average steering success of 70% on a set of articulated CALVIN tasks and outperforming goal-conditioning by 5.4x when steered with low-quality objectives. We also successfully steer an off-the-shelf real robot policy to express preference for particular objects and even create novel behavior. Videos and other visualizations can be found on the project website: https://dynaguide.github.io

1 Introduction

The rise of large datasets and expressive policies has enabled complicated skills in robots like folding clothes [3, 56], making sandwiches [45], and washing dishes [7]. When these large policies are deployed in the dynamic real world, we face the challenge of *steering* them to match the needs of a specific scenario. This means finding a subset of the policy's behaviors that are appropriate for the scenario [49]. The policy's complexity means that we ideally want to accomplish this steering without needing to retrain the policy, sample excessively from it, or anticipate all the possible steering during policy training.

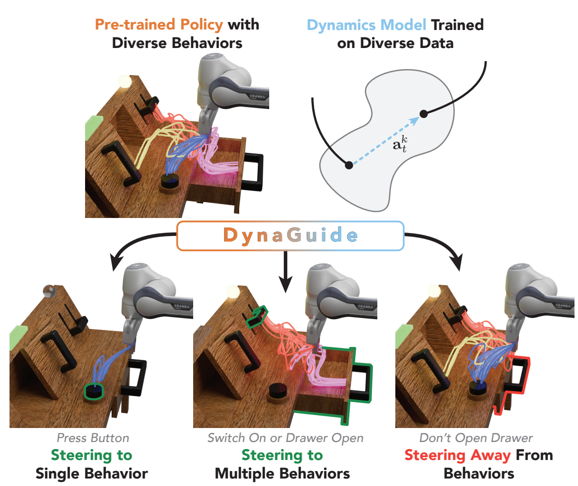

Figure 1: **DynaGuide** steers pretrained diffusion policies by adding guidance from a dynamics model into the action denoising process. This dynamics-based guidance can take a diverse behavior base policy and steer it towards one single behavior (left), multiple behaviors (middle), and even removing a behavior (right)— all without fine-tuning.

Most existing works for policy steering, however, rely on at least one of those assumptions. Even language or goal-conditioned policies are trained on a set of

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